Rahul Rawat

MACHINE LEARNING WORKING STUDENT

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on machine learning work spanning LLM based systems, rigorous model evaluation, and real time data pipelines. I have shipped a modular Retrieval Augmented Generation system with a custom decision making router on Llama 3.1 8b via Groq and LangChain, a fairness by design classification system validated to EU AI Act thresholds, and a recursive time series pipeline at eRay GmbH built on CatBoost multi quantile regression with strict anti leakage guarantees. Comfortable in Python, numpy, pandas, PyTorch style workflows, and Linux, I am the right fit for post training and evaluating robot foundation models.

EXPERIENCE

eRay GmbH
Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics
Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science
2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Hybrid RAG Orchestrator with Agentic Routing

Python | LangChain | Llama 3.1 8b via Groq | ChromaDB | HuggingFace MiniLM L6 v2 | Streamlit

- Delivered a modular Retrieval Augmented Generation system that classifies user intent into three execution paths, local knowledge retrieval, external web search, or direct conversational logic, measured by clean end to end responses across all three routes, using a custom decision making router on top of Llama 3.1 8b via Groq and LangChain orchestration.
- Built a persistent semantic memory over PDF and vector data, measured by consistent retrieval on cross session queries, using ChromaDB with local persistence and HuggingFace MiniLM L6 v2 embeddings.
- Kept multi turn context intact for the LLM, measured by coherent follow up answers rather than isolated single shot responses, by adding a stateful MemoryAgent directly into the inference pipeline.
- Shipped the whole system as a Streamlit interface with end to end ownership, measured by a working deployed prototype, using Groq inference and a modular routing layer that can be extended per new intent.

CreditIQ: Fairness by Design Credit Scoring

Python | scikit learn | AIF360 | SHAP | Streamlit

- Raised the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88, measured against the EU AI Act and AGG 80 percent fairness threshold, using AIF360 mitigation and threshold calibration on a real credit dataset.
- Diagnosed a hidden intersectional bias where younger women were still penalised even after single axis age bias was fixed, measured through SHAP driven subgroup analysis, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, measured on the held out test split, by documenting the fairness accuracy trade off as a deliberate and regulator defensible decision rather than a hidden compromise.
- Shipped a Streamlit decision support tool that gives finance managers a recommendation plus a plain language LLM generated explanation, measured against GDPR Article 22 and EU AI Act Article 14 human in the loop requirements, and backed the pipeline with unit tests at 100 percent branch coverage and a full regulatory write up.

Real Time Flight Tracking Data Pipeline

Python | PySpark | BigQuery | dbt | Apache Airflow | GCP Dataproc and GCS | Tableau with TabPy

- Collected live flight positions from the OpenSky Network API every 30 seconds and enriched them against airport, aircraft, and weather data across four sources, measured by a clean joined table covering more than 128 thousand records, using Python collectors and PySpark cleaning on Google Cloud.
- Shaped raw collector output into analysis ready tables with dbt and computed each aircraft's nearest airport along the way, measured by consistent nearest airport labels across the historical dataset, using PySpark for heavy lift and dbt for the modelling layer.
- Orchestrated the whole system with Apache Airflow so batch and real time layers refresh automatically every 15 minutes, measured by unattended DAG runs on GCS backed storage and Dataproc compute.
- Surfaced findings in a Tableau workbook backed by Python statistics through TabPy, measured by the finding that air traffic drops 4.4 times in heavy rain and clusters around hubs like Frankfurt and Munich, using dbt aggregates as the visual layer feed.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 patients, using 10 fold cross validation and per model confusion matrices to make the model comparison auditable.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance.

CERTIFICATIONS

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.